COMP 640 · EXPLAINABLE NLP · PROGRESS UPDATE

MemFaith

Measuring Causal Faithfulness of Memory Agents in Long-Context LLMs

Professor: Hanjie Chen

Students: Dev Sanghvi, Keyuan (Jade) Yan, Mohamad Kreidieh, Ansh Dabral

- **Evaluation Pipeline**
- **Counterfactual Data**
- **Inference Scaling**

SECTION 01

Keyuan (Jade) Yan

Counterfactual Data Engineering

Building a contamination-proof synthetic dataset for causal evaluation.

JADE · COUNTERFACTUAL DATA ENGINEERING

Counterfactual Dataset Generation

Counterfactual Dataset Generation

Developed a synthetic dataset generator utilizing fully fictional entities and facts (e.g., "Zarkov Lenn was a spectral dynamics researcher"). Ensures the LLM cannot rely on parametric memory from pre-training data.

Contamination Prevention

By using fictional knowledge, outputs are strictly driven by the provided context — not pre-training data. This guarantees a clean causal signal, free from knowledge leakage.

BM25 Distractor Retrieval

Implemented BM25-based distractor retrieval to pull highly confusing, structurally similar false segments — maximizing the difficulty of the causal discrimination task.

spaCy Preprocessing Pipeline

Built the spaCy-driven sentence chunking and context preparation scripts to ensure clean chunk boundaries for the ablation runner.

SECTION 02

Dev Sanghvi

Evaluation Pipeline & Baseline Proof

End-to-end causal ablation pipeline and baseline verification.

DEV · EVALUATION PIPELINE & BASELINE PROOF

Pipeline Finalization & Baseline Verification

Pipeline Finalization

Orchestrated the end-to-end causal ablation pipeline, generating the final combined_chunk_labels.csv containing over 11,200 labeled chunk rows — 30.6% causal, 69.4% non-causal — ready for discriminator training.

Chunking Algorithm Fix

Replaced greedy chunk-loading with an Agglomerative Merging approach, ensuring precisely balanced K-chunk distribution for stable and reproducible causality scoring.

The 'Context Overload' Experiment

Scaled context to 30 segments per example (1 gold evidence + 29 retrieved distractors) to maximize memory pressure and strictly fill a 1024-token window.

Baseline Verification & Key Finding

GPT-2 Small & XL failed entirely (100% No-Label rate). Migrated to Qwen2.5-0.5B-Instruct, which achieved exactly 33.1% accuracy — matching pure random chance on a 3-class task. Conclusion: heavy distractor context completely drowns out reasoning in small models, necessitating a context-pruning discriminator.

11,200+

Labeled Chunks

30.6%

Causal

69.4%

Non-Causal

33.1%

Baseline Accuracy (random chance)

SECTION 03

Mohamad Kreidieh

Discriminator Architecture, Training & Post-Pruning Evaluation

Coming soon — results to be added.

MOHAMAD · DISCRIMINATOR ARCHITECTURE & TRAINING

Results Pending

Insert updates regarding Discriminator Architecture, Training, and Post-Pruning Evaluation results here.

Discriminator Architecture

Details to be added

Training Methodology

Details to be added

Post-Pruning Evaluation Results

Details to be added

SECTION 04

Ansh Dabral

Inference Infrastructure & Scaling

High-throughput vLLM backend and HPC cluster orchestration.

ANSH · INFERENCE INFRASTRUCTURE & SCALING

vLLM Backend, Batch Orchestration & HPC Deployment

vLLM Backend Integration

Replaced the standard HuggingFace inference pipeline with a vLLM high-throughput backend, significantly accelerating causal ablation runs across large datasets. Enables production-scale inference speeds critical for iterating over 11,200+ labeled chunks.

Batch-Optimized Orchestration (BatchCCSRunner)

Implemented BatchCCSRunner, redesigning the evaluation loop to handle full-context predictions and iterative leave-one-chunk-out ablations in optimized, parallel batches — dramatically reducing wall-clock time per experiment.

HPC Deployment via SLURM

Developed SLURM submission scripts (run_all.sh) for seamless job orchestration on compute clusters. Enables automated, hands-off dataset scaling and model evaluation — fully reproducible across cluster environments.

Impact: The infrastructure built by Ansh enables the team to scale causal ablation experiments from dozens to thousands of examples — a prerequisite for training a robust discriminator.